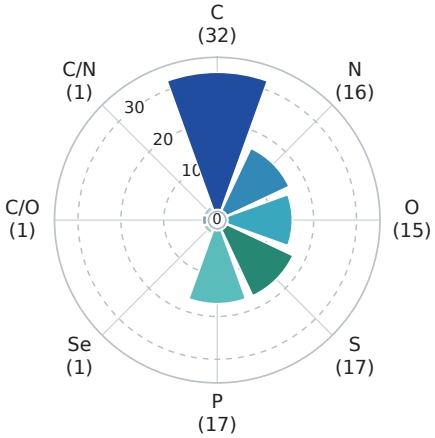
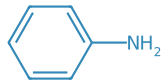
a Candidate diversity across reacting centres

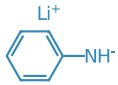


b Donor form expands the search

N–H and N salts

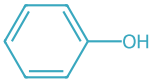


C048 Aniline

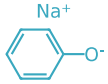


C045 Lithium anilide

O–H and O salts

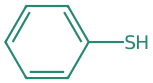


C051 Phenol

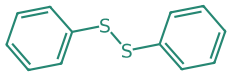


C064 Sodium phenoxide

S–H and S–S



C082 Benzenethiol



C078 Diphenyl disulfide

c Molecules retain alternative sites, pathways and endpoints

100
Candidate molecules

→

115
Site or pathway hypotheses

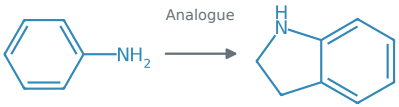
→

124
Proposed product options

d From recommended molecules to experimental donors

100 recommended molecules → **Expert assessment**
Structural relationships · Commercial accessibility · NP / NR
→ **100 final donors**
108 recorded input pairs

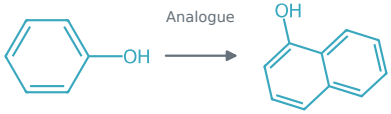
N–H model



C048 Aniline → AP039 Indoline

Actual-pair endpoint: NP2 / NR2

O–H model



C051 Phenol → AP048 1-Naphthol

Actual-pair endpoint: NP2 / NR2

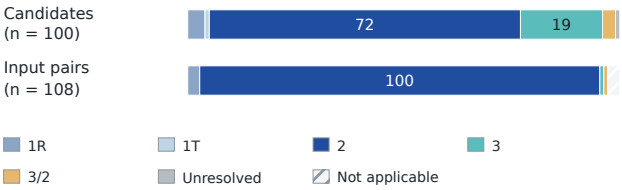
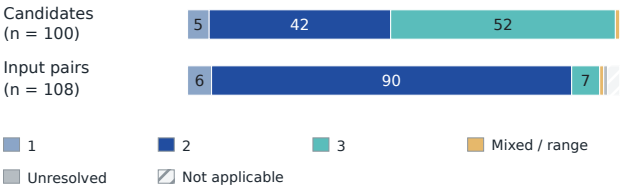
S–H model

C082 Benzenethiol → AP090 4-tert-Butyl-benzenethiol

Actual-pair endpoint: NP2 / NR2

e Novelty classifications by record unit

NP Product connectivity



NR Reaction precedent